

Is it a view or a copy?

Start from x, shown in the first row. For each operation `y = ...` below:

- 1. Try to fill in the metadata (`dtype`, `ndim`, `shape`, `strides`) that would be required to return `y` as a view of the same memory block as `x`. Can it be done?
- 2. Now decide: is `y` a view of `x`, or a copy? Base your answer on the outcome of step 1 and what we learned in the slides, then tick the box.

The one memory block behind `x` — 12 items of `int64`, 8 bytes each:

0	1	2	3	4	5	6	7	8	9	10	11
0	8	16	24	32	40	48	56	64	72	80	88

second row = byte offset of each item from the start of the block

NumPy operation	Array layout of y	NumPy array metadata	View or copy?																				
WORKED EXAMPLE x # the original array	<table><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>8</td><td>9</td><td>10</td><td>11</td></tr></table>	0	1	2	3	4	5	6	7	8	9	10	11	<table><tr><td>dtype</td><td>int64</td></tr><tr><td>ndim</td><td>2</td></tr><tr><td>shape</td><td>(3, 4)</td></tr><tr><td>strides</td><td>(32, 8)</td></tr></table>	dtype	int64	ndim	2	shape	(3, 4)	strides	(32, 8)	—
0	1	2	3																				
4	5	6	7																				
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dtype	int64																						
ndim	2																						
shape	(3, 4)																						
strides	(32, 8)																						
y = x.reshape((6, 2))	<table><tr><td>0</td><td>1</td></tr><tr><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td></tr><tr><td>6</td><td>7</td></tr><tr><td>8</td><td>9</td></tr><tr><td>10</td><td>11</td></tr></table>	0	1	2	3	4	5	6	7	8	9	10	11	<table><tr><td>dtype</td><td></td></tr><tr><td>ndim</td><td></td></tr><tr><td>shape</td><td></td></tr><tr><td>strides</td><td></td></tr></table>	dtype		ndim		shape		strides		☐ view ☐ copy
0	1																						
2	3																						
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8	9																						
10	11																						
dtype																							
ndim																							
shape																							
strides																							
y = x[:, :2]	<table><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>8</td><td>9</td><td>10</td><td>11</td></tr></table>	0	1	2	3	8	9	10	11	<table><tr><td>dtype</td><td></td></tr><tr><td>ndim</td><td></td></tr><tr><td>shape</td><td></td></tr><tr><td>strides</td><td></td></tr></table>	dtype		ndim		shape		strides		☐ view ☐ copy				
0	1	2	3																				
8	9	10	11																				
dtype																							
ndim																							
shape																							
strides																							
y = x[[1, 2, 0], [1, 1, 2]]	<table><tr><td>5</td><td>9</td><td>2</td></tr></table>	5	9	2	<table><tr><td>dtype</td><td></td></tr><tr><td>ndim</td><td></td></tr><tr><td>shape</td><td></td></tr><tr><td>strides</td><td></td></tr></table>	dtype		ndim		shape		strides		☐ view ☐ copy									
5	9	2																					
dtype																							
ndim																							
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y = x[[0, 2], :] # get the first and third row	<table><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>8</td><td>9</td><td>10</td><td>11</td></tr></table>	0	1	2	3	8	9	10	11	<table><tr><td>dtype</td><td></td></tr><tr><td>ndim</td><td></td></tr><tr><td>shape</td><td></td></tr><tr><td>strides</td><td></td></tr></table>	dtype		ndim		shape		strides		☐ view ☐ copy				
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second row = byte offset of each item from the start of the block

NumPy operation

Array layout of y

NumPy array metadata

View or copy?

y = x.ravel()

0	1	2	3	4	5	6	7	8	9	10	11
---	---	---	---	---	---	---	---	---	---	----	----

dtype	
ndim	
shape	
strides	

☐ view
☐ copy

y = x.T

0	4	8
1	5	9
2	6	10
3	7	11

dtype	
ndim	
shape	
strides	

☐ view
☐ copy

y = x.T.ravel()

0	4	8	1	5	9	2	6	10	3	7	11
---	---	---	---	---	---	---	---	----	---	---	----

dtype	
ndim	
shape	
strides	

☐ view
☐ copy